

LOGO

Quarterly System Analysis Report

Performance, reliability, and infrastructure review for Q4 2025

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1 Overview

1.1 Purpose

This report provides a comprehensive analysis of system operations during Q4 2025, covering service performance metrics, incident history, and infrastructure resource utilization. It identifies areas for improvement and proposes actionable next steps.

1.2 Scope

System	Description	Team
Web Service (API)	Customer-facing REST API server	Backend
Data Pipeline	Real-time data ingestion and transformation	Data Eng.
Monitoring	Infrastructure and service monitoring	SRE

1.3 Analysis Period

- **Start date:** October 1, 2025
- **End date:** December 31, 2025
- **Duration:** 3 months (Q4)

2 Detailed Analysis

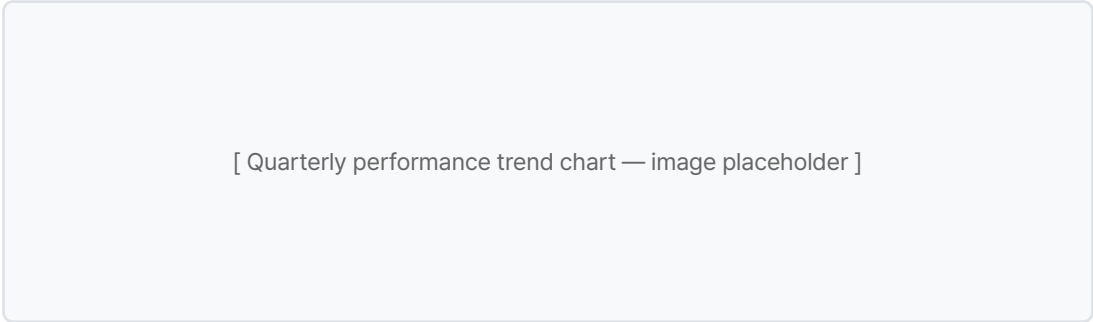
2.1 Service Performance Metrics

Service	Avg Latency	P99 Latency	Throughput	Uptime
API Gateway	45 ms	210 ms	12,500 req/s	99.97%
Auth Service	32 ms	185 ms	3,200 req/s	99.99%
Data Pipeline	120 ms	450 ms	8,700 msg/s	99.95%
Search Service	85 ms	320 ms	2,100 req/s	99.92%

2.2 Infrastructure Utilization

Resource	Allocated	Usage	Notes
CPU (vCPU)	256 cores	68%	Peaks at 85% during rush hours
Memory (RAM)	512 GB	72%	Includes cache
Storage (SSD)	10 TB	54%	Growing at 3% per month
Network	10 Gbps	35%	Stable

2.2.1 Performance Trend (Placeholder)



3 Key Findings

3.1 Performance Observations

The analysis revealed several noteworthy findings.

Performance Improvement

API Gateway average latency decreased by 15% compared to the previous quarter. This improvement is attributed to caching strategy optimization and connection pooling adjustments.

Warning: Storage utilization is growing at 3% per month. At the current rate, capacity will be exhausted within 6 months. A proactive expansion plan is recommended.

3.2 Action Items

The following items require confirmation or action from the relevant teams.

Question: The Search Service P99 latency was measured at 320 ms, but the previous quarterly report recorded 280 ms. Has the measurement methodology changed?

Q: Data Pipeline throughput drops significantly between 02:00–05:00. Is this caused by scheduled batch jobs?

Request: Please provide updated SLA targets for all services based on current operational standards.

Action: A storage expansion proposal should be prepared before the next operations review meeting.

4 Technical Details

4.1 Incident History

A total of 3 incidents occurred during the analysis period.

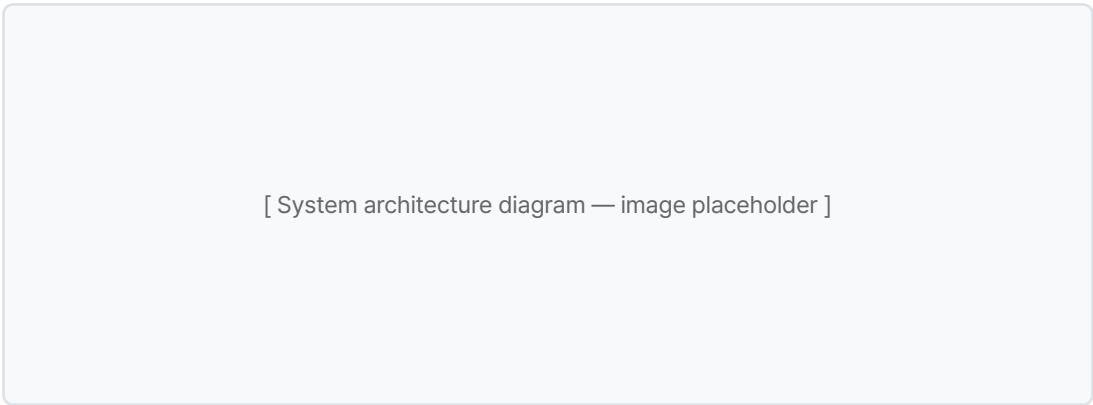
#	Description	Date	Recovery	Severity
1	API Gateway certificate expiry caused outage	10/15	23 min	High
2	DB connection pool exhaustion led to latency spike	11/08	12 min	Medium
3	DNS misconfiguration caused intermittent timeouts	12/22	45 min	Low

4.1.1 Root Cause Analysis

Each incident was traced to a specific root cause:

- Certificate expiry:** The auto-renewal script had a cron misconfiguration. `renewal_days` was set to `365` instead of `30`.
- Connection pool exhaustion:** Long-running batch jobs held connections, starving regular requests. Resolved by increasing `max_pool_size` from 50 to 100.
- DNS timeout:** Internal DNS cache TTL expired, causing upstream query delays. TTL was increased to 3600 seconds.

4.1.2 Architecture Diagram (Placeholder)



The architecture shown above reflects the current production configuration. A microservices migration is planned for Q1 2026. Updated architecture documentation will be distributed separately.

5 Summary

5.1 Conclusion

Overall system stability is satisfactory, with most services meeting SLA targets. However, storage capacity and Search Service latency require attention in the coming quarter.

5.2 Review Checklist

#	Item	Status
1	Search Service P99 measurement methodology change	To confirm
2	Nightly Data Pipeline throughput drop cause	To confirm
3	Updated SLA targets for all services	Requested
4	Storage expansion proposal	Action needed
5	Microservices migration schedule	To confirm

5.3 Next Steps

1. Develop storage expansion plan and request budget approval
2. Profile and optimize Search Service performance
3. Automate certificate renewal verification
4. Prepare for Q1 2026 microservices migration

5.4 Contact

For questions or feedback, please reach out to the team below.

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